

Intersection Of Three Planes

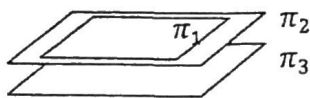
*****Major Step:** Determine how many normal vectors are parallel to each other***

Case 1: Three normals are parallel to each other. i.e. $\vec{n}_1 = k\vec{n}_2 = s\vec{n}_3$, $k, s \in \mathbb{R}$

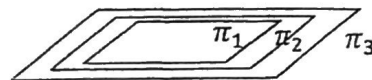
- Therefore, we have 3 parallel planes.



No pair of D values
are scalar multiples



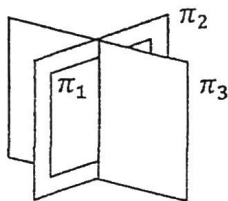
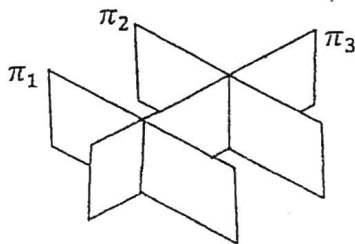
One pair of D values
are scalar multiples
 $D_1 = kD_2$



Two pairs of D values
are scalar multiples
 $D_1 = kD_2 = sD_3$

Case 2: Two normals are parallel to each other.

- Therefore, we have 2 parallel planes



Check to see if two planes are coincident.

- If not, you have the first picture
 - No Intersection
- If so, you have the second picture
 - Planes intersect in a line

Note: For the second picture:

$$\vec{n}_1 = k\vec{n}_2 \text{ and } D_1 = kD_2$$

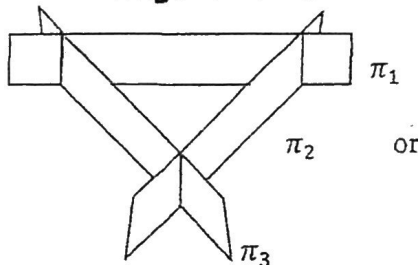
Case 3: No normals are parallel to each other

- Therefore, we have 0 parallel planes

→ Check if normals are coplanar i.e. Perform $\vec{n}_1 \cdot (\vec{n}_2 \times \vec{n}_3) = 0$??? or **Directly Solve**

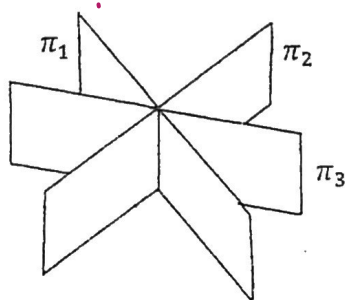
If normals are coplanar (i.e. $\vec{n}_1 \cdot (\vec{n}_2 \times \vec{n}_3) = 0$) you have:

Triangular Prism



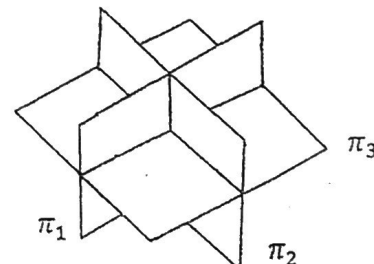
No intersection

Pinwheel



Line of intersection

If normals are not coplanar
(i.e. $\vec{n}_1 \cdot (\vec{n}_2 \times \vec{n}_3) \neq 0$) you have:



One point of intersection

→ You must solve the system to determine which case.

Strategy: Use ① and ② to get ④, then ① and ③ to get ⑤. Use ④ and ⑤ to solve for variables.

Find: 1) Contradiction ($0=3$) 2) True Statement ($0=0$) or 3) Unique values for variables

Intersection of 3 planes.

(1)

CASE I: 3 Parallel Planes.

* If $\vec{n}_1, \vec{n}_2$ & $\vec{n}_3$ are collinear, the planes are parallel & being distinct, do not intersect.

TYPE I: $\pi_1: 2x - 3y + z - 2 = 0$

$$\pi_2: 4x - 6y + 2z - 7 = 0$$

$$\pi_3: 6x - 9y + 3z + 15 = 0$$

$$\vec{n}_1 = [2, -3, 1] ; \vec{n}_2 = [4, -6, 2] ; \vec{n}_3 = [6, -9, 3]$$

$$D_1 = -2 ; D_2 = -7 ; D_3 = 15$$

$$\therefore \vec{n}_2 = 2\vec{n}_1 \quad \therefore D_2 \neq 2D_1$$

$$\frac{3}{2}\vec{n}_2 = \vec{n}_3 \quad \frac{3}{2}D_2 \neq D_3$$

$$\vec{n}_3 = 3\vec{n}_1 \quad D_3 \neq 3D_1$$

$$\therefore \pi_1 \parallel \pi_2 \parallel \pi_3$$

$\therefore$ 3 planes are distinct.

They do not intersect.

TYPE II:

(2)

$$\pi_1: 2x - 3y + z - 2 = 0$$

$$\pi_2: 4x - 6y + 2z - 7 = 0$$

$$\pi_3: 6x - 9y + 3z - 6 = 0$$

$$\vec{n}_1 = [2, -3, 1] \quad ; \quad \vec{n}_2 = [4, -6, 2] \quad ; \quad \vec{n}_3 = [6, -9, 3]$$

$$D_1 = -2$$

$$D_2 = -7$$

$$D_3 = -6$$

$$\therefore \vec{n}_2 = 2\vec{n}_1$$

$$\therefore D_2 \neq 2D_1$$

$$\frac{3}{2}\vec{n}_2 = \vec{n}_3$$

$$\frac{3}{2}D_2 \neq D_3$$

$$\vec{n}_3 = 3\vec{n}_1$$

$$D_3 = 3D_1$$

$$\therefore \pi_1 \parallel \pi_2 \parallel \pi_3$$

$\therefore \pi_1$ & π_3 are coincident

while π_2 is parallel & distinct.

$\therefore$ They do not intersect.

(3)

TYPE III : $\pi_1 : 2x - 3y + z - 2 = 0$

$\pi_2 : 4x - 6y + 2z - 4 = 0$

$\pi_3 : 6x - 9y + 3z - 6 = 0$

$\vec{n}_1 = [2, -3, 1] \quad , \quad \vec{n}_2 = [4, -6, 2] \quad ; \quad \vec{n}_3 = [6, -9, 3]$

$D_1 = -2 \quad , \quad D_2 = -4 \quad ; \quad D_3 = -6$

$\therefore \vec{n}_2 = 2\vec{n}_1 \quad D_2 = 2D_1$

$\frac{3}{2}\vec{n}_2 = \vec{n}_3 \quad \frac{3}{2}D_2 = D_3$

$\vec{n}_3 = 3\vec{n}_1 \quad D_3 = 3D_1$

$\therefore \pi_1 \parallel \pi_2 \parallel \pi_3$

π_1, π_2 & π_3 are $\parallel$ and coincident.

They intersect in the plane

$$2x - 3y + z - 2 = 0$$

CASE 2: 2 Parallel Planes.

(1)

* If only 1 pair of $\vec{n}_1, \vec{n}_2$ & $\vec{n}_3$ is collinear then there is a pair of parallel lines.

TYPE I: $\pi_1: 2x - 3y + z - 2 = 0$

$$\pi_2: 3x - 5y + 2z - 5 = 0$$

$$\pi_3: 6x - 9y + 3z - 7 = 0$$

$$\vec{n}_1 = [2, -3, 1] ; \vec{n}_2 = [3, -5, 2] ; \vec{n}_3 = [6, -9, 3]$$

$$D_1 = -2 \quad ; \quad D_2 = -5 \quad ; \quad D_3 = -7$$

$$k\vec{n}_2 \neq \vec{n}_1, k \in \mathbb{R} \quad \text{---}$$

$$t\vec{n}_2 \neq \vec{n}_3, t \in \mathbb{R} \quad \text{---}$$

$$\vec{n}_3 = 3\vec{n}_1 \quad D_3 \neq 3D_1$$

$\therefore \pi_1$ & π_3 are parallel and distinct.

π_1 & π_3 are parallel, π_2 is not parallel.

$\therefore$ They do not intersect.

TYPE II: $\pi_1: 2x - 3y + z - 2 = 0$

$\pi_2: 3x - 5y + 2z - 5 = 0$

$\pi_3: 6x - 9y + 3z - 6 = 0$

$\vec{n}_1 = [2, -3, 1]$; $\vec{n}_2 = [3, -5, 2]$; $\vec{n}_3 = [6, -9, 3]$

$D_1 = -2$; $D_2 = -5$; $D_3 = -6$

$k\vec{n}_2 \neq \vec{n}_1 \quad k \in \mathbb{R} \quad \text{---}$

$t\vec{n}_2 \neq \vec{n}_3 \quad t \in \mathbb{R} \quad \text{---}$

$\vec{n}_3 = 3\vec{n}_1 \quad D_3 = 3D_1$

$\therefore \pi_1$ & π_3 are parallel & coincident

π_2 is not parallel.

$\therefore$ they intersect at a line.

① Eliminate x

$3 \times \text{①} - 2 \times \text{②}$

$3(2x - 3y + z - 2 = 0)$
 $- 2(3x - 5y + 2z - 5 = 0)$

$y + z + 4 = 0$

$y = z - 4$

Let $z = t \Rightarrow$

② Eliminate y

$5 \times \text{①} - 3 \times \text{②}$

$5(2x - 3y + z - 2 = 0)$
 $- 3(3x - 5y + 2z - 5 = 0)$

$x - z + 5 = 0$

$x = z - 5$

$x = t - 5$
 $y = t - 4$; $z = t. \quad t \in \mathbb{R}.$

CASE 3 - No parallel planes but
normals are coplanar.

①

If the normals $\vec{n}_1, \vec{n}_2, \& \vec{n}_3$ are coplanar
but no pair is collinear then either the
planes have no point in common, or they
have a line in common.

TYPE I: $\pi_1: x + 2y + 3z - 4 = 0$

$$\pi_2: 2x - y + 4z + 7 = 0$$

$$\pi_3: 3x - 14y + z - 31 = 0$$

$$\vec{n}_1 = [1, 2, 3] ; \vec{n}_2 = [2, -1, 4] ; \vec{n}_3 = [3, -14, 1]$$

$$D_1 = -4 ; D_2 = 7 ; D_3 = -31$$

$$\left. \begin{array}{l} k\vec{n}_2 + \vec{n}_1 \\ + \vec{n}_3 \neq \vec{n}_2 \\ s \vec{n}_1 \neq \vec{n}_3 \end{array} \right\} \pi_1, \pi_2, \pi_3 \text{ are} \\ \text{all } \underline{\text{non-parallel}}.$$

check if co-planar:

$$\begin{aligned} & (\vec{n}_1 \times \vec{n}_2) \cdot \vec{n}_3 \\ & = [11, 2, -5] \cdot [3, -14, 1] \end{aligned}$$

$$= 33 - 28 - 5$$

$$= 0 \Rightarrow \text{the normals are coplanar.}$$

$$\begin{aligned} & \vec{n}_1 \times \vec{n}_2 \\ & = \begin{vmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \end{vmatrix} \\ & = [11, 2, -5] \end{aligned}$$

Eliminate x

$$2 \times (1) - (2)$$

$$2(x + 2y + 3z - 4 = 0)$$

$$- (2x - y + 4z + 7 = 0)$$

$$5y + 2z - 15 = 0 \quad (4)$$

$$4 \times (4) - (5)$$

$$4(5y + 2z - 15 = 0)$$

$$- (20y + 8z + 19 = 0)$$

$$-79 = 0$$

$$-79 \neq 0 \Rightarrow \text{no solution.}$$

(triangular prism)

No intersection.

Eliminate x

(2)

$$3 \times (1) - (3)$$

$$3(x + 2y + 3z - 4 = 0)$$

$$- (3x - 14y + z - 31 = 0)$$

$$20y + 8z + 19 = 0 \quad (5)$$

TYPE II: $\pi_1: 2x - y - z + 3 = 0$

$\pi_2: 2x - z + 1 = 0$

$\pi_3: 2x + y - z - 1 = 0$

$\vec{n}_1 = [2, -1, -1] ; \vec{n}_2 = [2, 0, -1] ; \vec{n}_3 = [2, 1, -1]$

$D_1 = 3 ; D_2 = 1 ; D_3 = -1$

$k\vec{n}_2 \neq \vec{n}_1 \quad k \in \mathbb{R}$

$t\vec{n}_3 \neq \vec{n}_2 \quad t \in \mathbb{R}$

$s\vec{n}_1 \neq \vec{n}_3 \quad s \in \mathbb{R}$

π_1, π_2, π_3 are all
non-parallel.

check if coplanar

$\vec{n}_1 \times \vec{n}_2 \cdot \vec{n}_3$

$= [1, 0, 2] \cdot [2, 1, -1]$

$= 2 - 2$

$= 0 \Rightarrow$ normals are coplanar.

$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} 2 & -1 & -1 \\ 2 & 0 & -1 \end{vmatrix}$

$= [1, 0, 2]$

Eliminate x

① - ②

$2x - y - z + 3 = 0$

$-(2x - z + 1 = 0)$

$-y + 2 = 0$

$y = 2$

Eliminate x

① - ③

$2x - y - z + 3 = 0$

$-(2x + y - z - 1 = 0)$

$-2y + 4 = 0$

$y = 2$

Sub. $y = 2$ into (1)

$$2x - y - z + 3 = 0$$

$$2x - 2 - z + 3 = 0$$

$$z = 2x + 1$$

Sub. $z = 2x + 1$ into (2)

$$2x - z + 1 = 0$$

$$2x - (2x + 1) + 1 = 0$$

$$0x = 0$$

$\Rightarrow$ true for all values of x

$\Rightarrow$ intersect in a line.

could let $x = t$
here:

let $z = t$

$$x = -\frac{1}{2} + \frac{t}{2}$$

$$y = 2.$$

$\Rightarrow$

$x = t$
$y = 2$
$z = 1 + 2t$

$$\therefore x = -\frac{1}{2} + \frac{t}{2}, y = 2, z = t.$$

CASE 3 : No planes parallel and the normals are not coplanar.

* If $\vec{n}_1, \vec{n}_2, \vec{n}_3$ are not coplanar then the planes intersect in a single point

TYPE III: $\pi_1: x + y - 3 = 0$
 $\pi_2: y + z + 5 = 0$
 $\pi_3: x + z + 2 = 0$

$\vec{n}_1 = [1, 1, 0] ; \vec{n}_2 = [0, 1, 1] ; \vec{n}_3 = [1, 0, 1]$

$k\vec{n}_2 \neq \vec{n}_1$
 $t\vec{n}_2 \neq \vec{n}_3$
 $s\vec{n}_3 \neq \vec{n}_1$

π_1, π_2, π_3 are non parallel.

check for coplanar

$\vec{n}_1 \times \vec{n}_2 \cdot \vec{n}_3$

$= \begin{vmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{vmatrix} \cdot [1, 0, 1]$

$= [1, -1, 1] \cdot [1, 0, 1]$

$= 1 + 1 = 2$

$\Rightarrow$ not coplanar.

Eliminate x

① - ③

$x + y - 3 = 0$

$-(x + z + 2 = 0)$

$y - z - 5 = 0$ (4)

④ - ②

$y - z - 5 = 0$

$-(y + z + 5 = 0)$

$-2z - 10 = 0$

$z = -5$

Eliminate y

① - ②

$x + y - 3 = 0$

$y + z + 5 = 0$

$x - z - 8 = 0$ (5)

⑤ + ③

$x - z - 8 = 0$

$x + z + 2 = 0$

$2x - 6 = 0$

$x = 3$

sub. $z = -5$ into (2)

$y + z + 5 = 0$

$y = 0$

$\therefore$ pt of intersection is at $(3, 0, -5)$